CRITICAL COMPLIANCE GAP

Deceptive Ad Vulnerabilities & Multi-Asset Contradictions

How unverified marketing creative leaks non-compliant discounts, uncertified claims, and invalidates brand truth.

$50,120	87%	5.4 Days	100%
FTC MAX FINE / VIOLATION	AD ASSET DRIFT RATE	AVG LEGAL BOTTLENECK	SHA-256 TAMPER PROOF

Existing Industry Problems (The Broken Status Quo)

1. Multi-Asset Contradiction Leaks

CRITICAL RISK

- **Desynchronized Channels:** Instagram caption claims 20% OFF while banner graphic declares 30% OFF.
- **Channel Leaks:** Tiered subscriber discounts mistakenly broadcast to general public ad feeds.
- **Silent Contradictions:** Disclaimers tucked away in terms PDFs contradict headline social ad copy.

2. Unsubstantiated Scientific Claims

REGULATORY FINE

- **Engineering Drift:** Copywriters state "24-hr cold retention" when lab certs only validate 12 hours.
- **Superlative Violations:** Unapproved words like "Miracle", "Cure", or "Guaranteed" evade manual eye checks.
- **Omission of Mandatory Qualifiers:** Missing required caveats ("While supplies last", "T&C Apply").

3. Velocity vs Legal Friction Bottleneck

OPERATIONAL LOSS

- **Review Gridlock:** Creative teams generate 50+ ad variants weekly; legal takes 4–7 days per audit.
- **Rushed Sign-offs:** 64% of brands bypass formal legal clearance under tight campaign launch deadlines.
- **Manual Fatigue:** High error rates when human reviewers manually cross-check PDF terms against images.

4. Post-Approval Silent Tampering

LEGAL LIABILITY

- **Out-of-Band Mod:** Creators edit approved copy or swap banner PNGs at 11 PM directly before go-live.
- **Lost Traceability:** Zero immutable record showing what was legally approved vs what was actually aired.
- **No Revocation Engine:** No system detects when an approved asset hash is modified post-sign-off.

The Core Governance Breakdown

FAILURE VECTOR	WHAT HAPPENS IN CURRENT WORKFLOW	REGULATORY & BUSINESS DAMAGE
Discrepant Discounts	Banner says 30%, checkout allows 20%, PDF says 15%	Consumer fraud complaints, chargebacks, brand trust erosion
Uncertified Claims	Marketing claims 24-hr vacuum lock without certified lab data	ASCI / FTC deceptive advertising investigations, mandatory recall
Banned Superlatives	Use of words like "Miracle Guaranteed" without qualification	Misleading advertising penalties, platform ad account bans
Last-Mile Edits	Asset modified in Canva after legal review approval	Brand counsel held legally liable for unauthorized creator copy

SYSTEM ARCHITECTURE

What We Fix: The 4-Layer Automated Trust Loop

Transforming subjective, manual ad reviews into deterministic mathematical compliance and cryptographic locks.

PILLAR 01

Single Source of Truth

- **Campaign Contract:** Canonical baseline of discount %, dates, approved claims, and banned terms.
- **Versioned Truth:** Immutable contract revisions (v1, v2) with audit provenance.

PILLAR 02

4-Layer AI Preflight

- **Automated Scan:** Audits social copy, banner images, and spec sheets simultaneously.
- **Contradiction Flags:** Instant grounding against active contract parameters.

PILLAR 03

Cryptographic Locking

- **SHA-256 Hash Seal:** Binds approved assets to contract version cryptographically.
- **Auto-Revocation:** Any post-approval tamper instantly revokes the seal to INVALIDATED.

The 4-Layer Inspection Protocol Breakdown

Layer 1: Deterministic Campaign Facts & Math

DETERMINISTIC

- **Discount Verification:** Matches caption and graphic numbers to contract offer % (e.g. 20% vs 30%).
- **Date & Expiry Alignment:** Flags expired end dates or mismatches between graphic and calendar.
- **Channel Variant Protection:** Checks if an exclusive VIP discount is leaking into public channels.

Layer 2: Cross-Asset Consistency Engine

MULTIMODAL AI

- **Caption vs Banner vs PDF:** Compares Instagram text with embedded banner text and PDF sheets.
- **Specification Verification:** Verifies advertised claims against technical engineering limits.
- **Desynchronization Prevention:** Flags when banner graphics say one thing and captions say another.

Layer 3: Brand Rules & Superlatives

LEGAL NLP

- **Prohibited Terms Detection:** Traps regulatory trigger words ("Miracle", "Cure", "Unbeatable").
- **Approved Claim Enforcement:** Ensures pre-approved marketing claims are used verbatim.
- **Required Disclaimers:** Enforces mandatory qualifiers ("While supplies last", "Selected models only").

Layer 4: Multimodal Vision & Accessibility

OCR & VISION

- **OCR Banner Extraction:** Extracts burned-in text and numbers inside PNG/JPEG graphics.
- **Visual Contrast Verification:** Tests text legibility against background brightness (WCAG AA).
- **Alt-Text & Transparency:** Generates semantic compliance tags for accessibility compliance.

How We Fix Issues in Real-Time: 1-Click Harmonization

Automated Reconciliation Engine

1-CLICK REMEDIATION

- **Evidence Grounding:** Shows exact ground truth rule from contract alongside contradictory creative text.
- **Suggested Replacements:** System suggests verified copy (e.g. rewrites "24hr cold" to "12hr lab-tested cold").
- **1-Click Accept:** Creator or reviewer accepts fixes with one click, immediately recalculating pack hash.

UI IMPLEMENTATION SNAPSHOTS

Production Interfaces: Hub & Contract Truth

Real system screen captures demonstrating modular topic segregation and baseline truth establishment.

1. The Campaigns Hub Directory (/dashboard)

LIVE SYSTEM VIEW

- **Topic Segregation:** Disassembles bloated single-page apps into clean directory cards with status pills.
- **Real vs Demo Separation:** Live client campaigns separated from pre-seeded benchmark demo scenario.

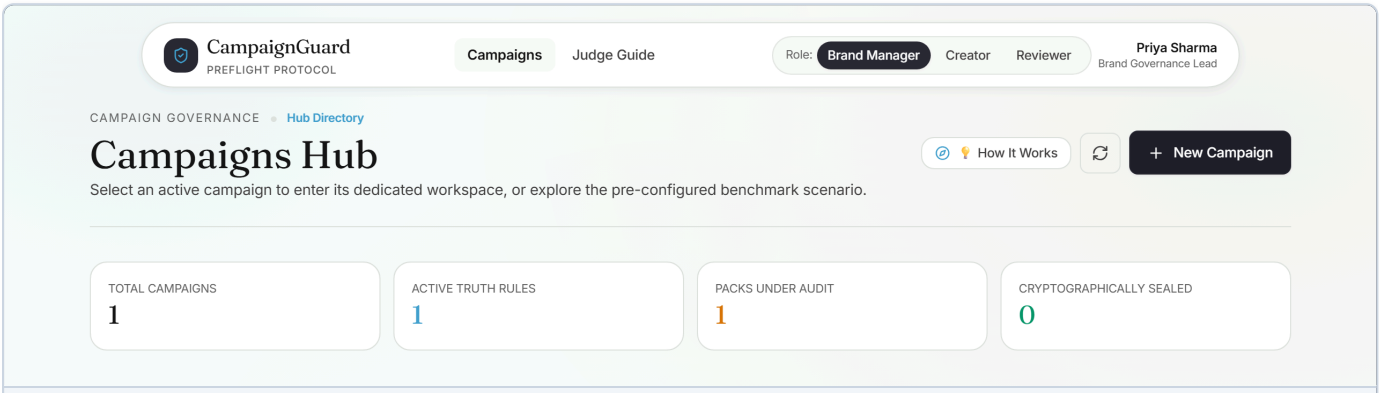


FIGURE 1.1: CampaignGuard Multi-Campaign Directory & Pre-Seeded Benchmark Hub cg.kshitijpro.xyz/dashboard

2. Step 1 • Truth Contract Builder (Brand Lead Persona)

STEP 1 • TRUTH BASELINE

- **Canonical Rules Definition:** Brand Lead sets discount %, expiry dates, approved claims, and prohibited terms.
- **Tiered Variants:** Sets channel-specific permissions (e.g. VIP subscribers allowed 30% discount).

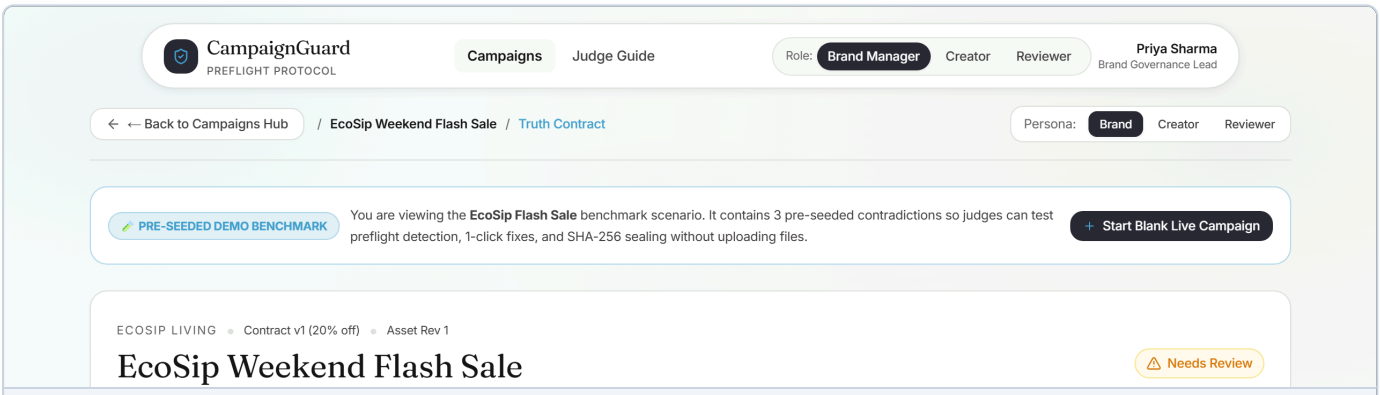


FIGURE 1.2: Campaign Contract Editor showing parameters, channel rules & version control cg.kshitijpro.xyz/dashboard?tab=contract

UI IMPLEMENTATION SNAPSHOTS

Production Interfaces: Creative Studio & AI Audit

Real system screen captures showing asset submission, automated detection, and cryptographic locking.

3. Step 2 • Creative Studio Pack Uploader (Creator Persona)

STEP 2 • CREATOR UPLOAD

- **Multimodal Ingestion:** Ingests social captions, banner graphic assets (PNG/JPEG), and PDF spec sheets.
- **Dynamic Hash Computation:** Generates unique cryptographic pack hash upon submission.

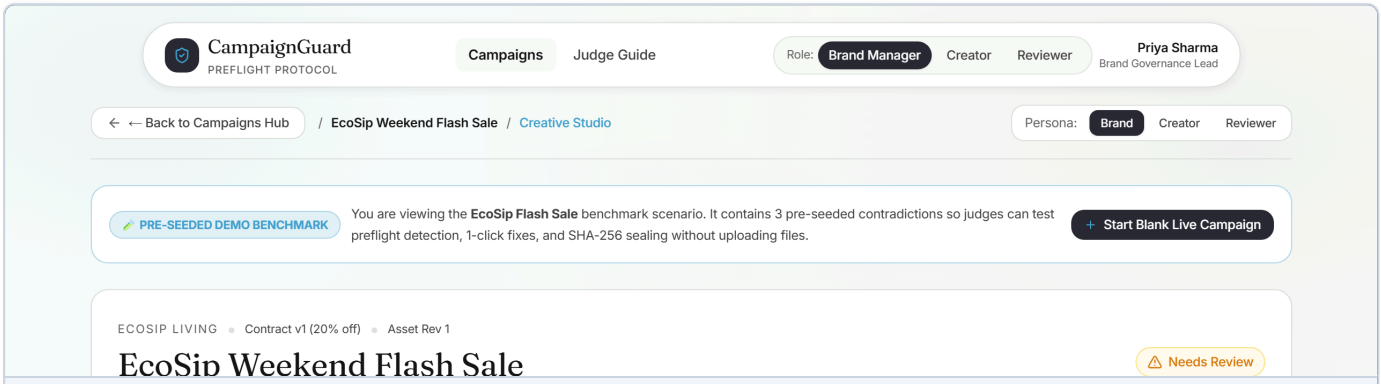


FIGURE 2.1: Creator Studio with caption input, banner uploader, and spec document ingestion cg.kshitijpro.xyz/dashboard?tab=creator

4. Step 3 • 4-Layer Preflight Inspection Findings Report

STEP 3 • AI CONTRADICTION SCAN

- **Clear Issue Cards:** Discloses contradiction headline, contract ground truth, and suggested replacement.
- **1-Click Remediation:** Single click accepts all recommendations and prepares campaign for reviewer lock.

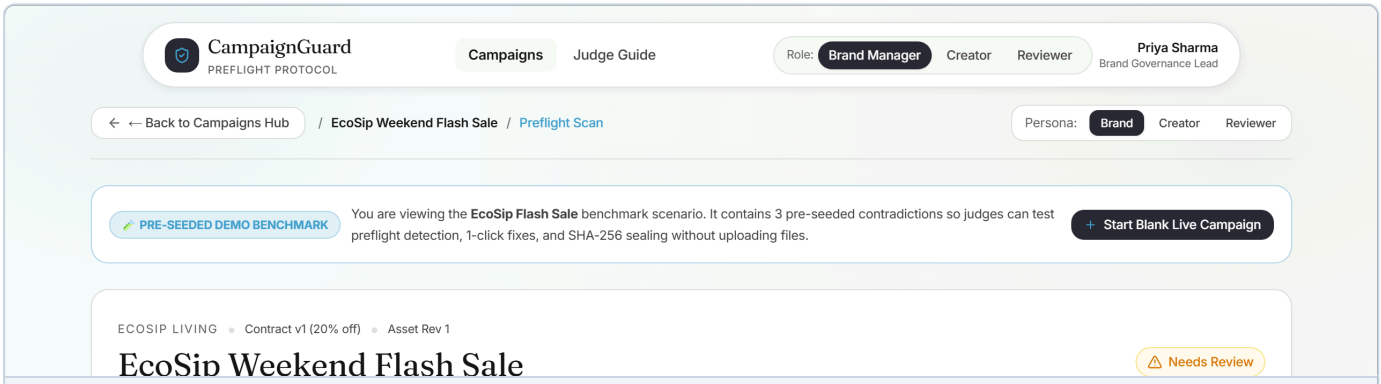


FIGURE 2.2: 4-Layer automated findings flagged across Facts, Cross-Asset & Brand Rules cg.kshitijpro.xyz/dashboard?tab=findings

TEST BENCHMARKS

Multiple Real-World Scenarios Tested

Real compliance failure simulations demonstrating CampaignGuard's automated detection and resolution.

SCENARIO 01

Flash Sale Discount Discrepancy & Channel Leak

LAYER 1: FACTS

Creative Input	Creator uploads banner stating "30% OFF Flash Sale" and caption stating "20% OFF".
Contract Truth	Contract v1 allows 20% discount for Public, and 30% ONLY for VIP Email Subscribers.
Engine Finding	**CONTRADICTION:** Banner claims 30% discount on public Instagram feed, exceeding 20% limit.
Remediation	System auto-corrects banner reference to 20% or suggests re-routing post to VIP channel.

SCENARIO 02

Scientific Exaggeration vs Engineering Spec Sheet

LAYER 2: CROSS-ASSET

Creative Input	Ad copy claims: "Miracle thermal lock guarantees 24 hours ice cold drinks for everyone!"
Contract Truth	Lab spec document certifies cold retention for up to 12 hours under laboratory conditions.
Engine Finding	**INSUFFICIENT EVIDENCE:** 24-hr claim lacks scientific basis; conflicts with 12-hr spec sheet.
Remediation	Copy rewritten to: "Keeps drinks cold for up to 12 hours based on certified thermal tests."

SCENARIO 03

Prohibited Superlatives & Omitted Legal Disclaimers

LAYER 3: BRAND RULES

Creative Input	Caption uses the prohibited word "Miracle" and lacks mandatory qualification disclaimer.
Contract Truth	Contract explicitly forbids "Miracle", "guaranteed", "cure-all"; requires "While stocks last".
Engine Finding	**PROHIBITED TERM DETECTED:** Word "Miracle" violates Section 43(a) trademark/ad regulations.
Remediation	Word "Miracle" stripped; mandatory qualifier "While supplies last on selected items" appended.

SCENARIO 04

Post-Approval Out-of-Band Asset Tampering

LAYER 4: CRYPTO REVOCATION

Attack Simulation	After Marcus Vance (Reviewer) locks pack with SHA-256 seal, creator edits text to 35% discount.
Crypto Engine	Recalculates SHA-256 payload hash of modified assets; compares with registered signature.
System Action	**INSTANT REVOCATION:** Campaign status switches from APPROVED to INVALIDATED.
Result	Distribution blocked, audit log records actor name, timestamp, and unauthorized payload.

GO-TO-MARKET STRATEGY

Target Markets, Stakeholder Personas & Approach

Defining who buys CampaignGuard, who uses it, and our frictionless enterprise land-and-expand strategy.

1. Targeted High-Stakes Market Segments

D2C & FMCG Consumer Brands

PRIMARY BUYER

- **Pain:** High-volume ad creative churn across Meta/TikTok with frequent discount desyncs.
- **Value:** Eliminates checkout discount errors, customer churn, and false advertising complaints.

Regulated Health, Wellness & Cosmetics

HIGH REGULATORY RISK

- **Pain:** Strict FDA, FTC, and ASCI guidelines regarding health claims and product superlatives.
- **Value:** Guaranteed adherence to lab-certified claims; prevents multi-million dollar class-actions.

FinTech, Banking & Credit Cards

MANDATORY DISCLOSURE

- **Pain:** Complex APR rates, cashback caps, and required statutory qualification disclaimers.
- **Value:** Ensures mandatory qualifiers are legally compliant across all visual formats.

Creator Agencies & Influencer Networks

HIGH VELOCITY

- **Pain:** Managing 200+ influencers who write their own rogue copy ignoring brand briefs.
- **Value:** Preflight verification tool given to creators before posts go live.

2. Key Stakeholder Personas & Value Delivered

USER ROLE	DAY-TO-DAY RESPONSIBILITIES	HOW CAMPAIGNGUARD EMPOWERS THEM
Brand Governance Lead *(e.g. Priya Sharma)*	Sets campaign claims, pricing tiers, legal qualifications, and channel variants.	Defines canonical Truth Contract once; rules automatically enforce across 1,000s of downstream assets.
Creative Copywriter / Agency *(e.g. Alex Chen)*	Drafts ad captions, designs banners, compiles promotional assets.	Gets instant AI preflight feedback; fixes issues in 1 click without waiting 5 days for legal email back-and-forth.
Authorised Reviewer / Legal *(e.g. Marcus Vance)*	Final gatekeeper signing off on promotional material.	Reviews verifiable evidence; seals approved assets with SHA-256 lock; auto-notified if copy is modified post-sign-off.

3. Enterprise Go-to-Market & Approach Strategy

PHASE 1: LAND

Agency Sandbox Entry

- Offer free preflight audits to top 50 influencer marketing agencies.
- Catch 10+ live compliance contradictions in their client campaigns to prove immediate ROI.

PHASE 2: INTEGRATE

Workflow Integrations

- Direct plugins for **Figma**, **Adobe Creative Cloud**, and **Canva**.
- Slack & Asana webhooks alerting legal teams only when unresolved contradictions exist.

PHASE 3: EXPAND

Enterprise Governance

- Annual software license sold to Brand VPs and Chief Legal Officers as regulatory insurance.
- Cryptographic audit ledger provided to corporate compliance officers during external audits.